

Multi-Modal, Training-Free Crack Extraction via Generalized FRANGI Graph

EUVIP 2026 – Luxembourg

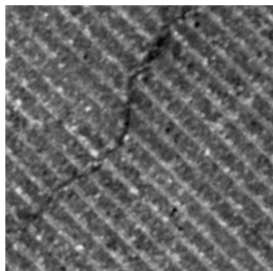
29 September 2026 | Paper 81

Louis HAUSEUX¹, Raphaël ANTOINE², Philippe FOUCHER²,
Pierre CHARBONNIER², Josiane ZERUBIA¹

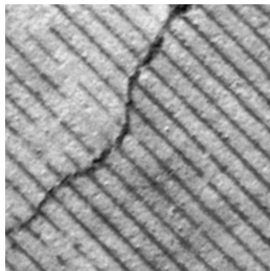
¹Inria, Université Côte d'Azur, Ayana research team, France

²Cerema, ENDSUM research team, France

Problem setting



Intensity



Range / depth

FIND dataset, aligned intensity and range views.

- ▶ Cracks are **thin**, **weak** and often **discontinuous**.
- ▶ Several modalities may observe the same geometry.
- ▶ Goal: extract an **editable crack skeleton without training**.

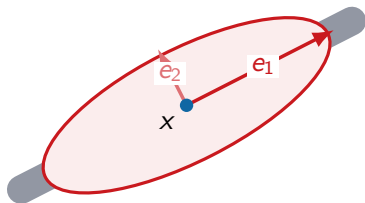
01

From pixels to a FRANGI graph



Classical FRANGI filter

$$\mathcal{H}_\sigma(x) = \nabla^2(G_\sigma * I)(x), \quad |\lambda_1| \leq |\lambda_2|.$$



► **Shape:** $R_B = |\lambda_1|/|\lambda_2|$

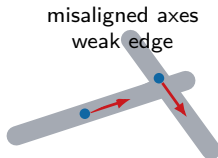
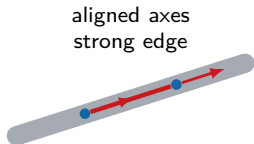
► **Contrast:** $S = \sqrt{\lambda_1^2 + \lambda_2^2}$

► **Direction:** e_1 follows the ridge

$$F_\sigma(x) = \exp\left(-\frac{R_B^2}{2\beta^2}\right) \left(1 - \exp\left(-\frac{S^2}{2c^2}\right)\right)$$

For $\lambda_2 > 0$; otherwise $F_\sigma = 0$.

Generalized FRANGI graph



- ▶ Pixels become **nodes**.
- ▶ Ridge compatibility becomes **edge similarity**.
- ▶ Alignment refines intensity and shape information.

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^{\sigma} S_{\text{shape}}^{\sigma} S_{\text{align}}^{\sigma}$$



Multimodal fusion

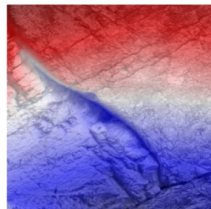
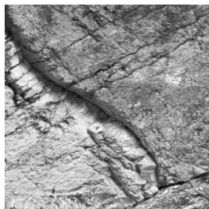
1. Normalize each Hessian

$$\hat{\mathcal{H}}_{\sigma}^{(m)}(x) = \frac{\mathcal{H}_{\sigma}^{(m)}(x)}{\max_{y \in \Omega} \|\mathcal{H}_{\sigma}^{(m)}(y)\|_2}$$

2. Fuse operators before eigendecomposition

$$\mathcal{H}_{\sigma}^{\text{fused}}(x) = \sum_m w_m \hat{\mathcal{H}}_{\sigma}^{(m)}(x)$$

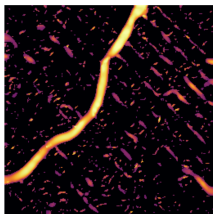
$$w_m \geq 0, \sum_m w_m = 1.$$



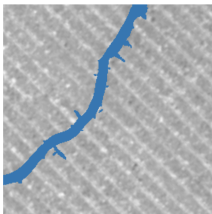
Intensity Depth overlay
Aligned modalities. © Cerema.



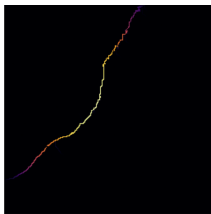
Extraction pipeline



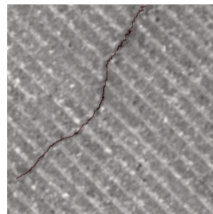
Similarity



Component



Centrality



Skeleton



$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}$$

$$\rho_{ij} = \|x_i - x_j\|_2 \leq R. \text{ [EUVIP 26]}$$

02

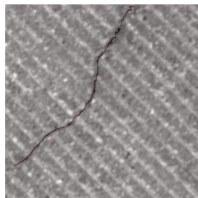
Experimental results



FIND: clean data



Ground truth



Our result

Intensity + range

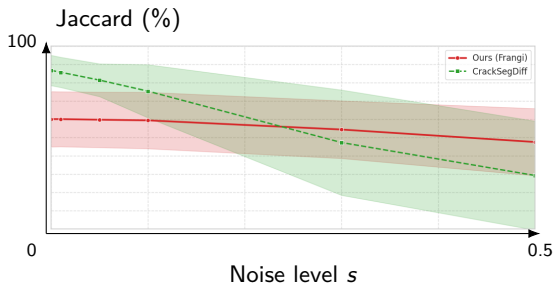
Method	IoU $\uparrow$	Tversky $\uparrow$	W $\downarrow$
Our graph	60%	68%	18 px
CrackSegDiff	87%	90%	5 px

Our ablation (IoU)

I	R	$I + R$	$I + R + F$
41%	54%	60%	63%

F : filtered range.

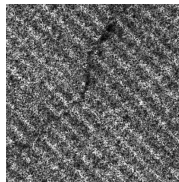
FIND: synthetic noise



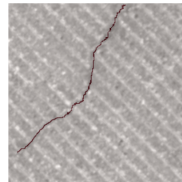
— Our graph — CrackSegDiff

$$I_s = I(1 + N), \quad R_s = R + N'$$

$N, N' \sim \mathcal{N}(0, s^2)$, independent.



Noisy intensity



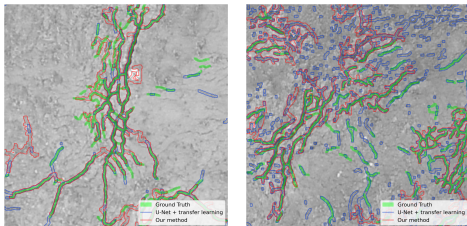
Our result, $s = 0.5$

- ▶ Lower initial accuracy
- ▶ Slower degradation in this test

Bands: standard deviation. [EUVIP 26; CrackSegDiff 24]

Transfer to geological images

Vaches Noires



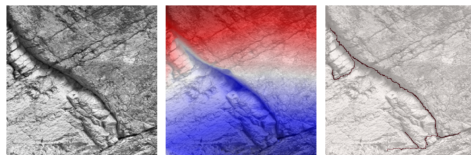
Test 1

Test 2

Ours Ground truth U-Net

IoU: 47 / 50% and 29 / 34% (ours / adapted U-Net).

Palais des Papes



Intensity

Depth overlay

Our result

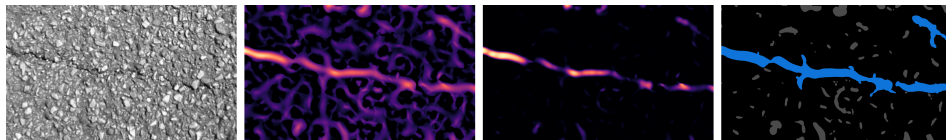
- ▶ Main fracture recovered
- ▶ No retraining

03

Beyond pairwise connectivity



Toward order $K = 2$



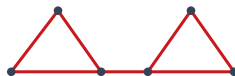
granular image

pixel-wise FRANGI

graph similarity

retained component

- ▶ Granular background can create spurious chains.
- ▶ $K = 1$: connectivity through vertices.
- ▶ $K = 2$: connectivity through triangles.



$K = 1$



$K = 2$

Conclusion

Main points

- ▶ FRANGI filter $\rightarrow$ FRANGI graph
- ▶ Multimodal fusion at the Hessian level
- ▶ Training-free extraction of crack skeletons

Next steps

- ▶ Evaluate higher-order connectivity
- ▶ Broader benchmarks
- ▶ Can a FRANGI graph hierarchy help guide SAM / CrackSAM?



[EUVIP 26; SAM 23; CrackSAM 24]

Bibliography: method and experiments

[Frangi 98]

A. F. Frangi, W. J. Niessen, K. L. Vincken, M. A. Viergever.
"Multiscale vessel enhancement filtering." MICCAI, 1998.

[GRETSI 25]

L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia.
"Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures au sein d'images d'un glissement de terrain. Comparaison avec une méthode d'apprentissage profond." GRETSI, 2025.

[EUVIP 26]

L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia.
"Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

[FIND 22]

S. Zhou, C. Canchila, W. Song.
"Fused Image dataset for convolutional neural Network-based crack Detection (FIND)." Zenodo, 2022. [Dataset record](#).

[CrackSegDiff 24]

X. Jiang, L. Jiang, A. Wang, K. Zhu, Y. Gao.
"CrackSegDiff: Diffusion Probability Model-based Multi-modal Crack Segmentation." [arXiv:2410.08100](#), 2024.



Bibliography: higher order and perspectives

[ANS 26]

L. Hauseux, K. Avrachenkov, J. Zerubia.
“Generalization of Single-Linkage with higher-order interactions.” Applied Network Science, 11, article 9, 2026.

[Thesis 26]

L. Hauseux.
“Utilisation de graphes pour la classification et l'extraction de structures. Généralisation à des interactions d'ordre supérieur.” PhD thesis and defense, Université Côte d'Azur, 8 September 2026.

[SAM 23]

A. Kirillov et al.
“Segment Anything.” ICCV, pp. 4015–4026, 2023. [Paper](#).

[CrackSAM 24]

K. Ge, C. Wang, Y. Guo, Y. Tang, Z. Hu, H. Chen.
“Fine-tuning vision foundation model for crack segmentation in civil infrastructures.” Construction and Building Materials, 431, 136573, 2024. [DOI](#).

Acknowledgments: Bpifrance / LiChIE (2020–2025); Université Côte d'Azur / DS4H (ANR-17-EURE-0004) and 3IA (ANR-19-P3IA-0002).